RAHMAH ABDULKARIM

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RESEARCH INTERESTS

Machine learning and statistical methods for clinical and medical imaging applications, with particular interest in how models behave when data is scarce, imbalanced, or shifts between training and deployment. Drawn to questions of model trustworthiness: how to measure it, stress-test it, and communicate it clearly in high-stakes settings where decisions affect patient outcomes.

EDUCATION

Rice University **Houston, TX**
M.A. Computational & Applied Mathematics & Operations Research, GPA: 3.76 *May 2026*
- **Relevant Coursework:** Modern Machine Learning, Bayesian Statistics, Stochastic Processes, Advanced Numerical Analysis, Natural Language Processing, High Performance Computing

Howard University **Washington, DC**
B.S. Mathematics; Minors: Computer Science, Chemistry, Latin; GPA: 3.91 *May 2024*
- **Relevant Coursework:** Data Structures and Algorithms, Real Analysis, Probability and Statistics II, Chemistry II & Lab, Organic Chemistry II & Lab, Physics I and II for Science and Engineering, Numerical Analysis

EXPERIENCE

Research Assistant **Aug 2024–May 2026**
Rice University, in collaboration with Houston Methodist *Houston, TX*
- Developed a multi-source clinical imaging integration and spatial-calibration pipeline across approximately 18,000 cardiac imaging files, linking source studies to echocardiography clips, model-generated segmentation overlays, and CMR references to construct a high-confidence analytical cohort.
- Reconstructed physically calibrated ventricular measurements from segmentation outputs and evaluated them against cardiac MRI references.
- Validated the revised workflow on held-out CMR references, reducing EF mean absolute error by 14.5% and RMSE by 15.3%.

Research Assistant **May 2023–Aug 2023**
Rice University, Google-Sponsored REU *Houston, TX*
- Converted 42,401 irregular seabed sediment observations into 15,345 validated nodes with terrain, collection, and spatial features for research on sediment characterization and seafloor mapping.
- Compared seven spatial-neighborhood definitions across 34 geographic blocks to quantify connectivity and coverage tradeoffs.
- Developed and validated an Extra Trees grain-size model that reduced mean RMSE by 16.2% and outperformed prior boosting in all five geographic folds.

Research Assistant **Feb 2023–Dec 2023**
Howard University *Washington, DC*
- Reconciled interview, examination, and laboratory records for 39,156 participants into a validated analytical cohort of 17,345 adults with 18 demographic, behavioral, and clinical predictors.
- Compared five models on 4,115 held-out adults; gradient boosting achieved 0.793 AUROC, 0.743 AUPRC, 89.5% sensitivity, and 15.8% lower Brier error than full logistic regression.
- Audited performance across 15 demographic groups, identifying an 11.2-point age-group AUROC gap and translating subgroup findings and limitations into research tables and figures.

Research Assistant **Aug 2022–Dec 2022**
Howard University, Department of Mathematics *Washington, DC*
- Reproduced and validated a MATLAB kinetic-component pipeline across three 20-frame [^{11}C]-(R)-PK11195 PET scans and 949,425 brain voxels, recovering archived component maps with 98.7%–99.99% agreement.
- Stress-tested reference-tissue extraction; regularization reduced held-out RMSE by 7.95% and maintained coefficient-map correlations of 0.995–0.999 under 10% controlled intensity perturbation.

SELECTED PROJECTS

RSNA Intracranial Aneurysm Detection

- Built a reproducible PyTorch pipeline for 14-label classification from 3D CTA studies, covering DICOM loading, Hounsfield-unit conversion, voxel resampling, cached tensors, class-imbalance handling, anatomically correct left–right augmentation, k-fold out-of-fold evaluation, 2D-to-3D ResNet weight inflation, TorchScript export, and inference benchmarking.

Football Scouting Intelligence System

- Built an end-to-end scouting system from StatsBomb event data, engineering 15 performance features across 88,513 player-match observations and aggregating them into 1,633 qualified player profiles for similarity-based replacement recommendations.
- Evaluated standardized Euclidean, cosine, five-component PCA, and five-dimensional autoencoder representations using neighborhood coherence and split-half and temporal stability; selected Euclidean retrieval after it achieved 66.3% role coherence@10 versus a 29.6% random baseline.
- Backtested 1,925 players across nine competition-seasons, recovering 36.7% of future top-10 neighbors and remaining 20.1% closer in held-out performance than same-role random candidates.
- Productionized an explainable recommendation engine through FastAPI, 20 automated tests, Docker, and GitHub Actions CI; deployed the API through Render and AWS ECR/ECS/Fargate and launched a public Streamlit dashboard.

Football Performance Intelligence

- Transformed 97,545 player-match records into 6,928 validated team-match observations and 144 competition-season profiles using SQL and Python.
- Developed a competition-relative passing-control index combining volume, completion accuracy, and progression; identified 16 high-control, low-threat teams and 17 efficient attacking benchmarks.
- Evaluated performance patterns across nine chronological holdouts and presented competition-relative comparisons through Tableau visualizations.

Premier League Manager Impact Analysis

- Prepared and validated 7,600 team-match observations representing 3,800 Premier League matches across ten seasons, with no missing values, duplicate records, incomplete match pairs, or manager-date violations.
- Developed hierarchical Bayesian Poisson models to estimate changes in goals scored and conceded during matches 1–3, 4–6, and 7–10 of a manager's tenure while controlling for home advantage, team strength, opponent strength, and season variation.
- Found no credible evidence of an immediate attacking new-manager bounce; teams were instead associated with 10.9% more goals conceded during matches 7–10 versus established-manager periods, with a 95% credible interval of 3.1%–19.0%.

Computational Linear Algebra

- Implemented tested dense matrix and vector abstractions in C++ and structure-aware low-rank and scaled block-diagonal matrices in Julia; benchmarked specialized matrix-vector multiplication at 11.8× and 98.8× the median speed of corresponding dense operations.

SKILLS & LANGUAGES

Statistical & ML Methods:	Bayesian modeling, statistical inference, uncertainty quantification, model evaluation, bias analysis, subgroup analysis, feature engineering
Programming & Tools:	Python (PyTorch, scikit-learn, pandas, NumPy, OpenCV), R, MATLAB, C++, SQL, Git, Docker, AWS, MLflow, Linux, Jupyter
Data Modalities:	Medical imaging (DICOM, CTA, echocardiography), structured clinical data, geospatial signal data
Instructional Tools:	Microsoft Office (Word, PowerPoint, Excel), Google Workspace, Jupyter Notebook

TEACHING EXPERIENCE

Independent Mathematics Tutor, Calculus **May 2025–Aug 2025**
Private *Houston, TX*

- Independently designed and delivered a full summer Calculus curriculum for junior high students, creating structured lesson plans from scratch tailored to each student's level and learning pace.
- Managed all aspects of instruction independently: lesson delivery, assignment creation, grading, and written feedback.
- Built foundational Calculus intuition in students well ahead of their grade level using concrete examples and incremental problem-solving.

Teaching Assistant, Department of Latin **Aug 2021–May 2024**
Howard University *Washington, DC*

- Designed and delivered lesson plans independently in the professor's absence; introduced varied instructional techniques to improve student retention and engagement.
- Proctored examinations, graded submissions, and provided detailed written feedback; held weekly office hours and recitation sessions.

Khan Academy Teaching Fellow **Sep 2023–May 2024**
Howard University *Washington, DC*

- Tutored 25 high school students in College Algebra, providing individualized one-on-one sessions aimed at reducing failure rates; graded assignments and provided detailed feedback.

Student Tutor, Algebra, Calculus, Pre-Calculus **Aug 2022–May 2023**
ACE Learning Support Services, Howard University *Washington, DC*

- Supported over 70 students using clear, structured problem-solving methods; identified gaps in understanding and adapted explanations to meet students at their level.

LEADERSHIP AND ACTIVITIES

- **First Year Representative, Muslim Students' Association:** Advocated for 15+ freshmen; planned 10+ events for a 400+ member community, Howard University (2021–22).
- **Secretary, CHANGO Afro-Latino Organization:** Maintained records, coordinated communications, and facilitated cultural programming, Howard University (2021–22).
- **Engineers Without Borders:** Led chapter communications and coordinated member updates, Howard University (2022–23).

VOLUNTEERING

Howard University Alternative Spring Break **March 2023**
New Orleans, Louisiana (60 hours)

- Tutored over 25 students in an after-school program.
- Led panels for high school students from underprivileged backgrounds on the importance of attending college.
- Participated in community service, including drainage cleanup in flood-prone areas.

PROFESSIONAL AFFILIATIONS & MEMBERSHIPS

Honor Societies:	Phi Beta Kappa, Eta Sigma Phi, Pi Mu Epsilon
Professional Organizations:	American Mathematical Society (AMS), Society of Women Engineers (SWE), Association for Computing Machinery (ACM)
Service Organizations:	Howard University Alternative Spring Break (HU ASB), Engineers Without Borders (EWB), Howard University Chapel, Muslim Student Association (MSA)

HONORS AND AWARDS

- **Capstone Scholar:** Awarded a $100,000 competitive scholarship based on demonstrated scholarly intellect, community service, and leadership.
- **Artishia & Frederick Jordan Scholar:** Awarded a $16,000 scholarship based on academic potential, good character, and community service.
- **Summa Cum Laude, National Latin Exam (2021):** Top 5% of exam takers nationwide.
- **Alain Locke Awardee:** Recognized for maintaining a perfect 4.0 GPA as an underclassman.
- Solveig Espelie Mathematics Scholar and Donald Cox Awardee.